

Designing Less Depth-Sensitive Crypto Perp Tests

Core judgment

A bar-based research stack is defensible for some crypto perpetual strategy families, but only if the expected edge lives at horizons slower than quote-by-quote microstructure, if liquidation is modeled on **mark** rather than last trade, and if fees, slippage, participation, and same-bar ambiguity are handled adversarially. That position follows directly from exchange mechanics: major venues compute mark from index, funding basis, and order-book inputs, and use mark for unrealized PnL and liquidation triggers rather than last-traded price. Binance, Bybit, OKX, and Hyperliquid all document this explicitly, though with venue-specific formulas and update cadences. ¹

The useful dividing line is not “trend versus mean reversion” by itself. It is whether the strategy’s realized edge depends mainly on **state variables observable in bars** or on **queue position, spread capture, transient order-flow imbalance, event latency, or first-minute dislocations**. Market-microstructure research shows that short-horizon price changes are strongly linked to order-flow imbalance and market depth, while crypto-specific work shows that order-book variation materially affects trading costs and that LOB data adds information beyond price series for short-horizon forecasting. Those findings are exactly why some families can be screened with 1m/5m bars plus mark/funding/OI, while others need book or trade replay before you should trust any Sharpe. ²

For your list, the clean split is:

- **Reasonably screenable and often Tier 1-validatable on liquid Tier A/B names:** liquid leader continuation, time-series momentum, sector leader continuation, risk-off swing shorts, and close-confirmed versions of prior-high momentum. These are slower, more state-based families; they do not inherently require queue priority to exist. ³
- **Conditionally screenable, but only with restrictive implementation rules:** post-breakout retest, post-catalyst base continuation, and D4-style deleveraging/reclaim longs. These can be screened with bars only if you avoid “touch” fills, avoid same-bar heroics, and push entries to confirmation rather than the exact wick level. Otherwise they become microstructure trades. ⁴
- **Not honestly serious-validated with bar data alone:** small-cap reversal, listing/event shorts, and ORB/funding-window strategies. These are precisely the families where short-horizon order flow, transient spreads, queue position, event timing, and venue-specific mechanics dominate realized fills. Funding-window trades are additionally tied to exchange-specific settlement schedules and interval changes; listing/delisting trades face special contract rules and automatic settlement behavior. ⁵

The practical consequence is simple: **Tier 1 is for family selection and conservative ranking, not for pretending you have execution solved**. If a family only works when you assume passive fills, same-bar target-first resolution, no mark-based liquidation, or bar participation large enough to lean on invisible depth, it is not a Tier 1 strategy. ⁶

Execution-sensitivity matrix

The matrix below answers the user's core question in operational form. "Serious validation" means enough evidence to decide whether a live canary is justified.

Strategy family	Primary execution sensitivity	Tier 0 OHLCV screen	Tier 1 1m/5m + mark/funding/OI	Serious validation tier	Conservative execution model that is still acceptable
Liquid leader continuation	Low to moderate on major names; edge usually lives over many bars, not single-book events. Major venues expose mark, index, funding, and OI histories suitable for this state-based testing. ⁷	Yes	Yes	Tier 1	Bar-close or next-bar entry, all-taker base case, low participation cap, mark-based liquidation.
Prior-high momentum	Moderate. A close-confirmed breakout is Tier 1-testable; a stop-entry through highs is much more microstructure-sensitive because the fill depends on the path through the bar and local order flow. ⁸	Limited	Yes, if close-confirmed	Tier 1 for close-confirmed; Tier 2 for touch/stop-entry variants	No touch fills on the breakout bar; execute next bar with stressed taker cost.

Strategy family	Primary execution sensitivity	Tier 0 OHLCV screen	Tier 1 1m/ 5m + mark/ funding/OI	Serious validation tier	Conservative execution model that is still acceptable
Time-series momentum	Low. This is the least depth-sensitive family on the list if restricted to liquid names and slower holding periods. Crypto TSMOM evidence is largely based on price series rather than L2 mechanics. ⁹	Yes	Yes	Tier 1	Next-bar execution, all-taker base, funding accrued exactly, low bar participation.
Sector leader continuation	Low to moderate. Main risk is cross-sectional ranking quality and universe churn, not queue dynamics, if you stay in liquid sector leaders. ¹⁰	Yes	Yes	Tier 1	Same as liquid leader continuation, with sector membership frozen ex ante.

Strategy family	Primary execution sensitivity	Tier 0 OHLCV screen	Tier 1 1m/ 5m + mark/ funding/OI	Serious validation tier	Conservative execution model that is still acceptable
Post-catalyst base continuation	Moderate to high. If the alpha is the base and continuation after the first reaction , Tier 1 can screen it; if the alpha is tied to news timestamp, exchange announcement latency, or immediate reaction , you need event replay. Listing/pre-market rules and funding exceptions matter. 11	No	Conditional	Tier 1 only after excluding reaction window; Tier 3 if catalyst timing is part of the signal	Exclude first reaction window; no same-bar continuation entries; use wide stress.
Post-breakout retest	Moderate to high. A retest defined on close/reclaim can be screened in Tier 1. A retest defined as “touch level, fill passively, bounce” is a queue-position trade and needs L2. 8	No	Conditional	Tier 1 for close-confirmed retests; Tier 2 for touch-limit versions	No passive-touch fill assumptions; fill only after close confirmation or next-bar open.
Risk-off swing shorts	Low to moderate if implemented on liquid names and holding over hours to days. Bars, funding, and OI are usually sufficient for first serious work. 12	Yes	Yes	Tier 1	Next-bar taker execution, mark-based risk, full funding accrual, low leverage.

Strategy family	Primary execution sensitivity	Tier 0 OHLCV screen	Tier 1 1m/ 5m + mark/ funding/OI	Serious validation tier	Conservative execution model that is still acceptable
Small-cap reversal	High. Crypto LOB studies show shallow books and rapidly rising liquidity cost with size, especially outside major names; short-term reversal in small/ illiquid coins is explicitly linked to low liquidity. ¹³	No	No, except idea triage only	Tier 2 or Tier 3	Bar tests are not decision- grade.
Listing/event shorts	Very high. Exchange-specific listing, pre-market, funding, and delisting rules change the economics; execution depends on event-time order flow and contract lifecycle. ¹⁴	No	No, except anecdotal triage	Tier 3	Full event replay, venue- specific lifecycle rules, and trade/ depth replay required.
D4-style deleveraging/ reclaim longs	High around the flush and reclaim, because liquidation cascades and reclaims are wick- and path-sensitive. Mark-based liquidation is central here; last-trade candles are not enough. ¹⁵	No	Conditional	Tier 1 for coarse post-reclaim screening; Tier 2 for real validation	Only trade after reclaim closes; model liquidation off mark extremes; heavy slippage stress.

Strategy family	Primary execution sensitivity	Tier 0 OHLCV screen	Tier 1 1m/ 5m + mark/ funding/OI	Serious validation tier	Conservative execution model that is still acceptable
ORB/funding-window strategies	High to very high. ORB lives in early-bar path dependence; funding-window trades live around known settlement times and venue-specific interval rules. Those are exactly the places where order flow and queue become decisive.	No	Exploratory only	Tier 2 for ORB, Tier 2 or Tier 3 for funding-window implementations	If screened at all, use very heavy cost stress and do not promote without book/trade data.

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A good internal rule is: **if performance changes materially when you replace “touch fills” with “close-confirmed fills,” the family was not Tier 1-validatable in the first place.** That rule follows from the intrabar-assumption problem documented in backtesting platforms and from the microstructure literature showing that short-horizon returns depend on order-flow imbalance and depth. 17

Conservative Tier One execution model

The fee floor for liquid centralized perp venues is not zero, and Tier 1 work should not assume it is. Current official schedules on major venues put non-VIP or simple baseline perp fees roughly around **4.5–5.5 bps taker per side** and **1.8–2.0 bps maker per side** on standard centralized venues, though the exact rate varies by venue, contract type, tier, promotions, and region. Binance documents USD/Bitcoin starting taker and maker rates of 4.5 and 1.8 bps in its current futures-fee explainer; Bybit’s VIP 0 perpetual/futures table shows 5.5 and 2.0 bps; OKX’s futures examples use 5.0 and 2.0 bps. 18

That leads to a simple default policy for Tier 1 on **liquid Tier A/B names**:

- **Base-fee model:** assume **all-taker** unless the strategy’s logic can survive that. For current centralized venues that means about **9–11 bps round trip in fees alone** before slippage. 18
- **Alternative sensitivity case:** if the strategy naturally rests orders but you do **not** have queue/trades data, run a second case with **mixed fills** but do **not** credit maker rebates as alpha. In other words, you may lower assumed fees somewhat, but you should not assume reliable passive capture without evidence. Maker/taker status depends on whether the order actually adds or removes liquidity, not on whether it was submitted as a limit order. 19
- **Funding model:** use the **actual venue funding history and actual settlement timestamps** whenever possible. Funding is exchanged only if the position is open at the settlement time, but the

interval itself is venue- and sometimes contract-dependent: Binance and OKX are eight-hour by default but can adjust to shorter intervals on selected contracts; Bybit can shift intervals under stress; Hyperliquid computes an 8h-equivalent rate but settles hourly. Cross-venue research therefore has to normalize rates to an 8h equivalent for comparison, while single-venue validation should simulate the exact venue timestamps. ²⁰

Family-specific Tier One stress assumptions

The table below gives **research defaults**, not universal truths. The purpose is to bias results against you unless the family is genuinely robust.

Family bucket	Eligible families	Extra round-trip slippage stress on top of fees	Participation cap for a small account	Notes
Low execution sensitivity	liquid leader continuation, time-series momentum, sector leader continuation, risk-off swing shorts	Base 4 bps / Hard 8 bps / Kill 12 bps	Max 0.50% of 1m notional or 1.00% of 5m notional	These caps keep you in a low-participation regime where bar testing is less dependent on hidden depth. Impact literature links cost to traded volume and participation; crypto liquidity work shows order-book variation directly affects PnL. ²¹
Moderate execution sensitivity	close-confirmed prior-high momentum, close-confirmed post-breakout retest	Base 6 bps / Hard 12 bps / Kill 20 bps	Max 0.25% of 1m or 0.50% of 5m	Breakout and retest entries cluster around visible levels, so crowding risk is higher. Use close-confirmation only in Tier 1. ⁸
Event-shaped but still screenable	post-catalyst base continuation after the base has formed	Base 8 bps / Hard 16 bps / Kill 25 bps	Max 0.25% of 1m or 0.50% of 5m	Exclude the initial reaction window from Tier 1. If the edge vanishes once the first reaction is excluded, it was an event-timing trade, not a bar-pattern trade. ²²
Deleveraging/reclaim screens	D4-style reclaim longs after reclaim close	Base 8 bps / Hard 16 bps / Kill 30 bps	Max 0.25% of 1m or 0.50% of 5m	Because liquidation and reclaim paths depend on mark and wick order, Tier 1 should only test delayed reclaim entries, never wick catches. ¹⁵

Family bucket	Eligible families	Extra round-trip slippage stress on top of fees	Participation cap for a small account	Notes
Depth-required families	small-cap reversal, listing/event shorts, ORB/funding-window	Not decision-grade in Tier 1	Not decision-grade in Tier 1	If you need these assumptions to make the backtest work, upgrade the data, not the story. ²³

Those participation caps are deliberately low. The square-root impact literature and more recent work on participation both say that impact grows with executed size and, for a given size, rises with participation. Crypto-specific LOB work also shows that outside the largest names, books are relatively shallow and large trades are timed when liquidity cost is lower. If your Tier 1 backtest needs more than a low single-digit share of the observed bar notional, you are already leaning on invisible depth. ²⁴

Data requirements by tier

A major reason Tier 1 is viable at all is that the core state variables are available directly from exchange infrastructure. Binance exposes klines, index-price klines, mark-price klines, funding history, and OI/OI statistics for perps. OKX exposes mark-price candles, index candles, funding channels, and open-interest channels. Bybit provides historical index price, mark price, and order-book downloads. Hyperliquid exposes candle snapshots and L2 snapshots through the API, but its historical S3 archive is explicitly incomplete and only guarantees L2 snapshots there, which means venue-specific collection matters. ²⁵

Minimum data for screening

Tier	Minimum data	What it is good for	What it is not good for
Tier 0 OHLCV screen	Trade-price OHLCV only	Idea triage, universe filters, very rough signal ranking on slow families	Any claim about real tradability, mark-based risk, funding-aware returns, or stop-path accuracy
Tier 1	1m and 5m bars, mark price series, index price series, funding history + timestamps, OI series, contract lifecycle/ listing/delisting calendar, venue fee tables, risk-tier/ maintenance-margin schedule	Conservative research on liquid, slower families; ranking strategy variants; kill/no-kill decisions before paying for L2	Queue position, passive fill realism, opening-range path dependence, event latency, small-cap reversals, listing/event trades

Tier	Minimum data	What it is good for	What it is not good for
Tier 2	Tier 1 + top-of-book and L2 depth snapshots , bid/ask quotes, ideally liquidations/order-book imbalance proxies	Validation of touch entries, passive logic, ORB-style timing, reclaim/retest implementations closer to reality	Sub-millisecond sequencing, exact trade-through replay, queue reconstruction without trade prints
Tier 3	Tier 2 + trade prints , event timestamps, announcement feed, event-replay plumbing, queue logic where relevant	Listing/event shorts, funding-window microstructure, short-horizon ORB, maker-dominant execution logic	Nothing important from this list remains hidden if the replay is implemented well

Minimum data for family-specific validation

Family	Minimum data for family-specific validation
Liquid leader continuation, time-series momentum, sector leader continuation, risk-off swing shorts	Tier 1
Prior-high momentum	Tier 1 if close-confirmed; Tier 2 if stop-entry or wick-triggered
Post-breakout retest	Tier 1 if retest and reclaim are close-confirmed; Tier 2 if touch-limit or maker-dependent
Post-catalyst base continuation	Tier 1 if the catalyst window is excluded and the signal begins after a visible base; Tier 3 if catalyst timing is part of alpha
D4-style deleveraging/reclaim longs	Tier 1 for delayed reclaim screens; Tier 2 for real validation
ORB/funding-window	Tier 2 minimum, Tier 3 preferred
Small-cap reversal	Tier 2 minimum, often Tier 3
Listing/event shorts	Tier 3

Minimum data for live canary

For a live canary, the minimum should be **at least one tier above the tier that made you believe the strategy**. In practice that means:

- Tier 1 families can go to canary with **Tier 1 historical validation plus live collection of quote/fill telemetry**, but only on the most liquid names and only with all-taker assumptions in the paper model until live evidence says otherwise. ¹⁹

- Conditional families should not go to canary unless you have **Tier 2 historical validation** and live quote/fill logging.
- Listing/event shorts and funding-window microstructure trades should not go to canary without **Tier 3-style replay data** and venue-lifecycle controls, because exchange rules around settlement, funding-interval changes, and delisting are part of the trade itself. ²⁶

Bar-backtest rules that reduce self-deception

Mark-price liquidation without full tick data

Do **not** model liquidation on last-traded-price candles. Major venues use mark for liquidation and unrealized PnL, and their mark can differ meaningfully from last because it incorporates index, funding basis, and order-book information. Binance states that liquidation is triggered when mark reaches liquidation price and that the actual liquidation-price print can differ because mark is not linear through a volatile move; Bybit says the same in its dual-price and liquidation docs; Hyperliquid also uses mark for liquidations and TP/SL triggers. ²⁷

A conservative no-tick model is:

1. Revalue the open position each bar using **mark**.
2. For a **long**, test liquidation against the **bar low of mark**; for a **short**, test against the **bar high of mark**.
3. If the liquidation threshold is breached anywhere in the bar, treat the position as liquidated **before** discretionary exits on that bar.
4. Realize the closeout at the **more adverse of the liquidation threshold and the bar extreme**, then add the venue's taker fee and an extra liquidation penalty buffer. This is intentionally harsh, but it is consistent with exchange documentation saying mark triggers liquidation and realized closeout can differ from the displayed trigger. ²⁸

If a strategy dies under that rule, the backtest was leaning on favorable hidden path assumptions.

Same-bar ambiguity

Intrabar ambiguity is not a cosmetic issue. TradingView's own documentation says its emulator must make assumptions about intrabar movement on historical bars unless lower-timeframe inspection is used; the Bar Magnifier exists specifically because those assumptions change simulated fills. ²⁹

For Tier 1, the robust rule set is:

- **Signals are generated on bar close only.**
- **New entries never fill on the same bar that created the signal.**
- If a prior resting order and a stop/target are both hit in the same bar, resolve in the **adverse order**.
- If both stop and target are touched in the same bar, assume the **stop** is hit first.
- If liquidation threshold is touched in the same bar as a target, assume **liquidation first**.
- Track the **ambiguous-trade rate**. If it is not trivial, the family is telling you it needs lower-timeframe data or L2.

That policy is pessimistic, but it is much closer to honest than letting the engine guess a favorable bar path.

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One-minute versus five-minute bars

The difference between 1m and 5m is not just noisiness. It is how much unobserved path you have hidden inside each decision interval. Lower-timeframe intrabar inspection exists precisely because coarse bars force fill assumptions. More generally, order-flow and LOB effects matter most at very short horizons, which means 5m can hide the exact behavior that makes a strategy untradeable. 17

A workable rule:

- Use **5m** for slower continuation and swing families where the intended hold is many 5m bars and stops are not razor-tight.
- Use **1m** for anything involving breakout/retest timing, reclaim logic, funding windows, tight stops, or holds measured in the first few hours.
- Treat the **1m–5m delta itself as a stress test**. A real Tier 1 family should degrade between 1m and 5m; it should not flip from good to bad purely because the bar hid too much path.

Maker versus taker assumptions

Maker/taker is an execution outcome, not a wish. Binance and OKX both define taker flow as immediate execution against resting book liquidity; a limit order can still be taker if it crosses or matches the book. 19

So the Tier 1 policy should be:

- **Default all fills to taker.**
- Only run a maker sensitivity case if the family can logically rest orders away from the market.
- Even in that case, **do not book expected maker rebates** unless you have evidence of passive fill rates from quote/trade data or live telemetry.
- Any family whose edge disappears when you switch assumed maker fills to taker fills is not ready for canary.

Mandatory stress tests

The minimum mandatory set is:

- Re-run with **2× slippage** and **all-taker fees**.
- Re-run with **half the participation cap**.
- Re-run on **1m and 5m**.
- Re-run with **adverse same-bar resolution**.
- Re-run with **mark-based liquidation** and tighter maintenance-margin assumptions.
- Re-run with **lifecycle exclusions** around listing, pre-market, delisting, and contract changes.
- Re-run **venue by venue** rather than pooling across exchanges, because crypto perp markets remain fragmented and transaction-cost-adjusted opportunities can reverse quickly. 30

If the family only survives in one relaxed simulation mode, that is not robustness. That is a story.

Evidence required before live canary

Backtest overfitting is not an abstract warning. Bailey and coauthors show that backtest overfitting is endemic once many configurations are tried, and the Deflated Sharpe Ratio exists precisely because selection bias and non-normal returns inflate reported performance. White's Reality Check and later model-confidence procedures exist because testing many variants on the same data can produce false discoveries. Separate evidence from Quantopian's cohort of live-traded algorithms showed that commonly reported backtest metrics, especially plain Sharpe, had little power to predict out-of-sample results on their own. ³¹

Backtesting validation checklist

Before a live canary, a Tier 1-eligible family should clear all of the following:

- Pass **walk-forward** and locked **out-of-sample** tests.
- Keep positive expectancy after **all-taker fees** and your hard slippage stress.
- Show acceptable performance on **both 1m and 5m**, or else explain why one timeframe is the true decision clock.
- Show low sensitivity to **same-bar resolution**.
- Use **mark-based** liquidation and venue funding timestamps.
- Keep trade-level participation under the hard cap in nearly all historical cases.
- Show that PnL is not concentrated in a few event days or a single venue.
- Report **trial count**, **Deflated Sharpe Ratio**, and either **White Reality Check / SPA** or an **MCS-style** multiple-model correction.
- Report an **ambiguous-trade rate** and **breakeven slippage**.
- Reconcile results across at least two venues if the contract exists on both, because price discovery and funding integration are incomplete across platforms. ³²

Live-canary checklist

A live canary should be tiny, boring, and diagnostic:

- Restrict to **Tier 1-eligible families on the most liquid names**.
- Start with **all-taker expectations** even if you submit some passive orders.
- Enforce **hard participation caps** lower than the backtest caps.
- Log **exchange clock**, **signal time**, **order-submit time**, **ack**, **partial fills**, **cancel/replace**, **funding debits/credits**, **mark at entry/exit**, and **realized slippage versus model**.
- Abort or cut size if realized slippage exceeds modeled slippage by a preset multiple over a small sample.
- Record **contract lifecycle events**, funding-interval changes, and any venue-rule notices.
- Run a daily reconciliation of realized fills and PnL against the Tier 1 simulator using the same mark/funding conventions.
- Do not enable maker-credit assumptions in production models until canary logs prove the passive fill rate and adverse-selection profile are acceptable. ³³

The evidence threshold before canary is not “the backtest looks good.” It is “the strategy still looks worth testing after we remove the backtest’s favorite lies.”

Backtesting agent prompt

You are implementing a Tier 1 research backtester for liquid crypto perpetual futures.

Your job is to produce conservative, execution-aware tests for strategy families that are eligible for Tier 1 validation with bar, mark, funding, OI, and lifecycle data.

Scope

- Allowed families:
 - liquid leader continuation
 - prior-high momentum, but ONLY as close-confirmed breakout
 - time-series momentum
 - sector leader continuation
 - post-catalyst base continuation, but ONLY after the initial reaction window has been excluded
 - post-breakout retest, but ONLY as close-confirmed retest/reclaim
 - risk-off swing shorts
 - D4-style deleveraging/reclaim longs, but ONLY after reclaim confirmation
- Disallowed for decision-grade testing in this engine:
 - small-cap reversal
 - listing/event shorts
 - ORB/funding-window strategies
 - any touch-entry, wick-catch, or maker-dependent implementation

Universe rules

- Use only liquid Tier A/B centralized-perp names.
- Exclude contracts with active pre-market status, special funding rules not captured in the input, imminent delisting, or missing mark/funding/OI history.
- Freeze the tradable universe using only information known at the evaluation time.

Required inputs per symbol

- Trade-price OHLCV bars at 1m and 5m
- Mark-price OHLC (or at minimum mark close and bar high/low proxies)
- Index-price OHLC or close
- Funding history with exact timestamps and interval metadata
- Open interest series
- Contract lifecycle metadata:
 - listing time
 - pre-market / innovation-zone / special-status flags if available
 - delisting / settlement notices if available
- Venue fee assumptions:
 - taker fee per side
 - maker fee per side
- Risk model inputs:

- maintenance margin schedule or a conservative approximation
- leverage / position sizing rules
- Optional but useful:
 - realized volatility
 - sector map
 - BTC / ETH regime filter
 - venue identifier

Core execution policy

- Generate signals using only information available at bar close.
- Never fill a newly generated trade on the same bar that created the signal.
- Default entry and exit fills to taker.
- Run all models in BOTH 1m and 5m.
- Primary fill price:
 - next bar open
 - then apply signed slippage and fees
- If the strategy explicitly enters on the close, allow close execution only if the design states "trade on close"; otherwise use next-open execution.

Transaction-cost policy

- Base case fees: all-taker
- Mixed maker/taker case: optional sensitivity only; do NOT assume maker rebates create alpha
- Slippage stress buckets, added on top of fees:
 - low-sensitivity families: 4 / 8 / 12 bps round trip
 - moderate-sensitivity families: 6 / 12 / 20 bps round trip
 - reclaim/event-shaped families allowed in this engine: 8 / 16 / 30 bps round trip
- If symbol is Tier B rather than Tier A, multiply slippage assumptions by 1.5x unless exact venue-specific calibration is provided
- Funding:
 - accrue only if the position is open at the funding timestamp
 - simulate exact funding timestamps from the input data
 - use the actual signed funding rate from the venue data
 - normalize to 8h-equivalent only for reporting and cross-symbol comparison, not for actual PnL simulation

Participation and capacity policy

- Hard cap order size by bar notional:
 - low-sensitivity families:
 - $\leq 0.50\%$ of 1m bar notional
 - $\leq 1.00\%$ of 5m bar notional
 - moderate-sensitivity families:
 - $\leq 0.25\%$ of 1m bar notional
 - $\leq 0.50\%$ of 5m bar notional
- If an order exceeds the cap:
 - either clip the order to the cap and record the clip
 - or reject the trade, depending on test mode

- Report capacity utilization for every trade

Mark-based risk and liquidation

- Use mark price, not last trade price, for unrealized PnL and liquidation checks
- For a long:
 - use the bar LOW of mark for worst-case intrabar margin testing
- For a short:
 - use the bar HIGH of mark for worst-case intrabar margin testing
- If liquidation threshold is breached anytime in the bar:
 - liquidation overrides discretionary exits on that bar
 - realize the exit at the more adverse of:
 - liquidation threshold
 - mark bar extreme
 - then add taker fees and an additional liquidation penalty buffer
- Record liquidation events separately from normal exits

Same-bar ambiguity policy

- Adversarial resolution is mandatory
- Rules:
 - if both stop and target are touched in the same bar, assume stop first
 - if both liquidation and target are touched in the same bar, assume liquidation first
 - if a prior resting order and an exit are both touched, assume the adverse sequence
- Report:
 - ambiguous-trade count
 - ambiguous-trade fraction
- If ambiguity materially affects results, flag that the strategy is not robust for Tier 1

Family-specific implementation rules

- liquid leader continuation:
 - allow bar-close trend filters and next-bar entries
- prior-high momentum:
 - breakout must be confirmed by close
 - no stop-entry-through-high fills
- time-series momentum:
 - allow slower momentum definitions and volatility scaling
- sector leader continuation:
 - rank within sectors using only ex-ante information
- post-catalyst base continuation:
 - exclude the initial catalyst reaction window from entries
 - only trade after a visible base and close-confirmed continuation
- post-breakout retest:
 - retest must be confirmed by close/reclaim
 - no touch-limit fills
- risk-off swing shorts:

- allow regime filters, index weakness, leader breakdowns
- D4-style deleveraging/reclaim longs:
 - require reclaim confirmation
 - no wick-catch entries

Validation and reporting

For every strategy run, output:

- net performance before and after costs
- performance under each slippage bucket
- 1m vs 5m comparison
- venue-by-venue comparison
- trade count
- turnover
- average holding period
- ambiguous-trade fraction
- liquidation count
- fraction of trades clipped by participation caps
- breakeven slippage
- funding contribution
- concentration metrics:
 - top 5 days contribution
 - top symbol contribution
 - top venue contribution
- robustness flags:
 - FAIL if edge disappears under all-taker + hard slippage
 - FAIL if 1m and 5m disagree in sign without a strong mechanical reason
 - FAIL if ambiguity is high
 - FAIL if capacity usage is persistently near the cap
 - FAIL if performance depends on pre-market, listing, delisting, or event windows excluded by policy

Research hygiene

- No look-ahead leakage
- No survivorship-biased universe
- No same-bar discretionary miracles
- No maker-rebate alpha without evidence
- No last-price liquidation modeling
- No pooling of venues into one synthetic fill stream
- When in doubt, choose the more adverse fill and the more restrictive interpretation

Final output format

- strategy summary
- assumptions used
- eligibility verdict:
 - Tier 1 acceptable
 - Tier 1 screen only
 - upgrade to Tier 2

- upgrade to Tier 3
- performance table across stress scenarios
- diagnostics table
- clear explanation of what would be required before live canary

The right standard here is not perfection. It is refusing to turn missing depth data into invented capacity and invented fills. On crypto perps, that is the line between a useful screen and a polished self-deception.

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